



# Working with Polynomials

A polynomial is a monomial or sum of monomials.

The prefix *poly-* means “many.” A polynomial can be a sum of many monomials.

## Definitions

A **monomial** is a number, a variable, or the product of a number and one or more variables. All variables in a monomial must have whole-number exponents. A monomial cannot have a variable in a denominator or under a radical symbol. Monomials are sometimes called terms.

The **coefficient** of a monomial with multiple factors is the numerical factor.

A monomial with no variable factors is called a **constant**.

A **polynomial** is a monomial or sum of monomials.

## Determining Whether an Expression Is a Polynomial

### EXAMPLE 1

Determine whether the expression is a polynomial. If it is not, explain.

**A**  $3\sqrt{x} + 17x + 1$

### SOLUTION

This expression is not a polynomial. It has a variable under a radical symbol.

**B**  $2x^3 - x^2 - 5$

**SOLUTION**

This expression is a polynomial. It is a sum of monomials because

$$2x^3 - x^2 - 5 = 2x^3 + (-x^2) + (-5).$$

**C**  $x^{-5} + 4x^4 - 1$

**SOLUTION**

This expression is not a polynomial. It has a variable with a negative exponent.

► **Think About It** In Example 1A,  $\sqrt{x} = x^{\frac{1}{2}}$ , so a variable has an exponent that is not a whole number.

In Example 1C,  $x^{-5} = \frac{1}{x^5}$ , so there is a variable in a denominator.

**D**  $\frac{2}{3}x^3 - x^0$

**SOLUTION**

This expression is a polynomial. Note that 0 is a whole number.

$$\frac{2}{3}x^3 - x^0 = \frac{2}{3}x^3 - 1 \quad \blacksquare$$

► **Remember** The whole numbers are  $\{0, 1, 2, 3, \dots\}$ .

## Definitions

**Like terms** are terms that contain the same variable factors taken to the same powers.

The **degree of a monomial** is the sum of the exponents of its variable factors.

The **degree of a polynomial** is the degree of the monomial with the greatest degree.

A polynomial is in **simplified form** when it has no like terms, every term is in simplest form, and its terms are in order of decreasing degree with respect to a variable.

The **leading coefficient** of a polynomial in simplified form is the coefficient of the first term.

## Monomials and Like Terms

### EXAMPLE 2

Which of the following are like terms?

$$6x^2 \quad x^2y \quad -2xy \quad -2x^2y \quad 10x^2 \quad -x^2$$

#### SOLUTION

$6x^2$ ,  $10x^2$ , and  $-x^2$  are like terms.

$x^2y$  and  $-2x^2y$  are like terms.

$-2xy$  is not like any of the other terms. ■

### EXAMPLE 3

Determine the degree of the monomial.

$$-21x^2y^5z^3$$

#### SOLUTION

Add the exponents of the variables.

$$-21x^2y^5z^3$$

$$2 + 5 + 3 = 10$$

The degree of the monomial is 10. ■

## Simplifying and Classifying Polynomials

### EXAMPLE 4

Simplify  $8x + 2x^3 + 9x + 2 - 6x^2 - 5x^3$ . Then identify the leading coefficient.

#### SOLUTION

Combine like terms.

$$8x + 2x^3 + 9x + 2 - 6x^2 - 5x^3$$
$$17x - 3x^3 + 2 - 6x^2$$

Then write the terms in order of decreasing degree. The simplified form is  $-3x^3 - 6x^2 + 17x + 2$  and the leading coefficient is  $-3$ . ■

► **Think About It** Combining like terms uses the distributive property.

$$8x + 9x = (8 + 9)x = 17x$$

A polynomial can be classified by its number of terms or by its degree.

## Classifying Polynomials by Number of Terms

| Type      | Number of terms | Example                               |
|-----------|-----------------|---------------------------------------|
| monomial  | 1               | 12<br>$-\frac{2}{5}xy^3$              |
| binomial  | 2               | $a + b$<br>$1 - 3c^5$                 |
| trinomial | 3               | $x^2 + 3x - 10$<br>$a^2 + 4ab + 4b^2$ |

► **Think About It** Prefixes can help you remember the number of terms in monomials, binomials, and trinomials.

**Mono-** means “one,” as in **mono**cle or **mono**poly.

**Bi-** means “two,” as in **bi**cycle or **bi**lingual.

**Tri-** means “three,” as in **tri**cycle or **tri**athlon.

Many of the polynomials in the study of algebra are polynomials in one variable.

## Definition

A **polynomial in  $x$**  is a polynomial of the form

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

where the exponents are all whole numbers, the coefficients are all real numbers, and  $a_n \neq 0$ .

Note that the polynomial in the definition can be written as

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x^1 + a_0 x^0.$$

The exponents are  $n, n-1, n-2, \dots, 2, 1$ , and  $0$ . The coefficients are  $a_n, a_{n-1}, a_{n-2}, \dots, a_2, a_1$ , and  $a_0$ .

## Classifying Polynomials by Degree

| Type          | Degree | Example                                           |
|---------------|--------|---------------------------------------------------|
| constant      | 0      | $-7$                                              |
| linear        | 1      | $9x - 12$                                         |
| quadratic     | 2      | $x^2 + 8x + 16$                                   |
| cubic         | 3      | $6x^3 - 2x^2 - 5$                                 |
| quartic       | 4      | $5x^4 - 3x + 1$                                   |
| quintic       | 5      | $-x^5 + 2x^4 - 3x^3 + 7x^2 - 10x + 1$             |
| $n$ th degree | $n$    | $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ |

### EXAMPLE 5

Classify  $5a^4 + a^2 - 3$  by its number of terms and by its degree.

#### SOLUTION

The expression  $5a^4 + a^2 - 3$  has three terms, so it is a trinomial. The term with the greatest degree is  $5a^4$ , and its degree is 4, so the degree of the polynomial is 4. It is a quartic trinomial. ■

### EXAMPLE 6

**A** Add horizontally.

$$(7x + 14) + (2x^2 - 3x + 2)$$

#### SOLUTION

$$\begin{aligned} (7x + 14) + (2x^2 - 3x + 2) &= 7x + 14 + 2x^2 - 3x + 2 \\ &= 2x^2 + 7x - 3x + 14 + 2 \\ &= 2x^2 + 4x + 16 \end{aligned}$$

Eliminate parentheses.

Write in order of decreasing degree.

Combine like terms.

**B** Add vertically.

$$(-4x^3 + 2x^2 + 9) + (7x^3 + 10x^2 - 3x - 4).$$

**SOLUTION**

$$\begin{array}{r} -4x^3 + 2x^2 \quad + 9 \\ + 7x^3 + 10x^2 - 3x - 4 \\ \hline 3x^3 + 12x^2 - 3x + 5 \end{array}$$

Write one polynomial under the other so that like terms appear in the same columns.

Add and combine like terms. ■

### EXAMPLE 7

**A** Subtract horizontally.

$$(a^2 + 3a) - (2a^2 - 5a + 1)$$

**SOLUTION**

To subtract polynomials, add the opposite of every term in the subtrahend.

The rest of the work is the same as adding polynomials.

$$\begin{aligned} (a^2 + 3a) - (2a^2 - 5a + 1) &= a^2 + 3a - 2a^2 + 5a - 1 \\ &= a^2 - 2a^2 + 3a + 5a - 1 \\ &= -a^2 + 8a - 1 \end{aligned}$$

► **Remember** minuend – subtrahend = difference

**B** Subtract vertically.

$$(2x^4 - 10x^2 - 6x + 1) - (2x^4 + 7x^3 + 6x + 1)$$

**SOLUTION**

$$\begin{array}{r} 2x^4 \quad \quad -10x^2 - 6x + 1 \\ - (2x^4 + 7x^3 \quad \quad + 6x + 1) \\ \hline \end{array}$$

Write the second polynomial under the first polynomial so that like terms appear in the same columns.

$$\begin{array}{r} 2x^4 \quad \quad -10x^2 - 6x + 1 \\ + (-2x^4 - 7x^3 \quad \quad - 6x - 1) \\ \hline -7x^3 \quad -10x^2 - 12x \end{array}$$

To subtract, add the opposite of every term in the second polynomial. ■

► **Think About It** Subtraction is not commutative. You must subtract the second polynomial from the first polynomial.